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Abstract*—Pneumonia remains a leading cause of morbidity and mortality worldwide, necessitating early and accurate diagnosis for effective treatment. While deep learning models, particularly Convolutional Neural Networks (CNNs), have shown promise in pneumonia detection from chest X-ray images, challenges such as overfitting and suboptimal model performance persist. This study proposes an optimized deep learning approach leveraging Genetic Algorithms (GA) to enhance the performance of the VGG16 model for pneumonia detection. By replacing traditional backpropagation with GA-based optimization, we aim to mitigate overfitting and improve classification metrics. The methodology involves fine-tuning the final layer of a pretrained VGG16 model using GA, where candidate models undergo mutation and selection based on fitness criteria such as accuracy, precision, recall, and F1-score. Comparative analysis with other optimization techniques, including Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Bayesian Optimization, demonstrates the efficacy of GA in achieving robust performance. Experimental results on a publicly available dataset reveal that the GA-optimized model achieves competitive accuracy while maintaining generalizability, highlighting its potential for real-world clinical applications. This research underscores the significance of metaheuristic optimization in advancing deep learning models for medical image analysis.

***Keywords*—Pneumonia, Deep learning, Optimized Genetic Algorithm, VGG16, Optimization, Pre-trained Model**

I. Introduction (*Heading 1*)

Pneumonia is an acute infection of the lungs that causes the alveoli to fill with fluid or pus [1] [2] . It is a leading cause of morbidity and mortality worldwide. Globally, pneumonia is the single largest infectious killer of young children and the fourth leading cause of death across all ages [1] [3] . According to WHO and UNICEF reports, over 800,000 children under age five died of pneumonia in 2017 [4] . One recent estimate notes that pneumonia claimed roughly 2.5 million lives in 2019 (about one person every 13 seconds), including 672,000 children under five [3] . These figures underscore the urgent public health burden of pneumonia, which disproportionately affects the very young, the elderly, and vulnerable populations. In pneumonia, pathogens (bacteria, viruses or fungi) inflame the lungs, filling air sacs with inflammatory fluid, which impairs gas exchange and causes symptoms like cough, fever, chest pain, and difficulty breathing [2] [3] . Early and accurate diagnosis is critical, since bacterial pneumonia can be treated with antibiotics, but delayed or missed diagnoses can lead to severe complications and death

Literature Review:

The optimization of deep learning models has become a crucial step to enhance performance, reduce computational costs, and improve generalization. Traditional training methods, such as stochastic gradient descent (SGD), although effective, can struggle with local minima and hyperparameter sensitivity. As a result, researchers have turned to metaheuristic algorithms such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Bayesian Optimization, Grey Wolf Optimizer (GWO), and Ant Colony Optimization (ACO) to optimize deep learning models.

Genetic Algorithms, inspired by the principles of natural evolution, have shown promise in evolving neural networks by adjusting weights, selecting optimal architectures, and fine-tuning hyperparameters. Bedi et al. (2024) proposed EvoLearn, a method that integrates GA with traditional backpropagation to optimize deep learning weights, showing significant performance improvements over standard training methods. Similarly, Jeong et al. (2022) introduced selective layer tuning in pretrained models such as VGG16 and ResNet using GAs, demonstrating that intelligently choosing which layers to fine-tune can yield higher accuracy with reduced computational effort.

In the context of deep convolutional neural networks (CNNs), López-Rincón et al. (2018) successfully employed GA to optimize hyperparameters for cancer classification tasks, outperforming manually tuned networks. Lee et al. (2021) further extended GA applications by evolving both the structure and hyperparameters of deep neural networks, emphasizing the flexibility and robustness of GA in high-dimensional search spaces.

Comparative studies between GA and other optimization techniques have provided deeper insights into their relative strengths. Purnomo et al. (2024) compared GA, ACO, and Harmony Search for CNN hyperparameter optimization, finding that although ACO achieved slightly higher accuracy, GA converged faster. Similarly, Lankford and Grimes (2024) showed that PSO often outperforms ACO in complex architecture search tasks, but GA remains competitive due to its diverse exploration capabilities.

Bayesian Optimization, particularly Gaussian-process-based models, has also been widely adopted for hyperparameter tuning. However, Raiaan et al. (2024) noted in their systematic review that metaheuristic approaches like GA, GWO, and ACO perform commendably in specific domains such as medical imaging and time-series forecasting, where the search space is complex and non-convex.

The Grey Wolf Optimizer (GWO) and Ant Colony Optimization (ACO) have gained attention as well, particularly for their collective behavior mechanisms which balance exploration and exploitation. Kaveh and Mesgari (2023) reviewed various metaheuristics and highlighted that while swarm-based methods (e.g., PSO, GWO) often converge quickly, evolutionary strategies like GA provide a better balance between global search and local refinement.

In summary, the literature suggests that Genetic Algorithms are highly effective for optimizing deep learning models, particularly when combined with pretrained architectures. The ability to evolve weights, selectively fine-tune layers, and adaptively search the solution space makes GA a strong contender against other metaheuristic and Bayesian methods. However, the choice of optimization technique often depends on the specific problem domain, model architecture, and computational constraints.

II. Related Work

In a recent study, researchers proposed a Convolutional Neural Network (CNN)-based model for detecting pneumonia using chest X-ray images to accelerate diagnosis and initiate treatment earlier. Due to the limited availability of original X-ray samples, the study employed a Deep Convolutional Generative Adversarial Network (DC-GAN) to generate synthetic images. The proposed model was evaluated and compared against pretrained VGG16 and ResNet50 models across three datasets: synthetic, original, and a combined dataset. The results showed that all models achieved their highest performance on the synthetic dataset, while the combined dataset—augmented with synthetic images—consistently outperformed the original dataset. Reported average test accuracies across the datasets were 93% (synthetic), 89% (original), and 91% (combined). Among the tested models, their custom CNN achieved the best results with 95%, 91%, and 93% accuracy, respectively. This study highlights the value of high-quality synthetic medical images in enhancing the effectiveness of deep learning models and addressing the data scarcity problem in AI-powered diagnostic systems.

[1] B. Atbaş, E. Calşşkan, and O. Erdem, "AI-Supported Pneumonia Diagnosis: Performance of Deep Learning Models and Synthetic Data Generation," in *Proc. 2024 Signal Processing: Algorithms, Architectures, Arrangements, and Applications (SPA)*, Poznan, Poland, Sep. 2024, pp. 1–6, doi: 10.23919/SPA61993.2024.10715621.

In another recent study, researchers proposed a pneumonia detection approach using chest X-ray images and the Xception CNN model combined with transfer learning. The study aimed to enhance diagnostic accuracy by leveraging transfer learning methods, which are especially effective for medical image classification. Using a dataset of 3450 chest X-ray images and training the model over 200 epochs, they applied data augmentation techniques such as Color Jitter to improve model generalization. The model achieved an impressive accuracy of 97%, demonstrating its reliability and potential as a decision support tool for medical practitioners in diagnosing pneumonia.

[2] M. Singla, K. S. Gill, S. Malhotra, and S. Devliyal, "Chest X-ray Image Pneumonia Detection using Transfer Learning and Xception CNN Model," in *Proc. 2024 International Conference on Intelligent Systems for Cybersecurity (ISCS)*, Gurugram, India, May 2024, pp. 1–6, doi: 10.1109/ISCS61804.2024.10581194.

A recent study by Singla et al. focused on improving the accuracy of pneumonia detection using chest X-ray (CXR) images by leveraging transfer learning and the Xception CNN model. The study employed a dataset of 3450 CXR images and trained the model over 200 epochs. To improve generalization, data augmentation techniques, particularly Colour Jitter, were applied. The model was evaluated using standard performance metrics such as precision, recall, F1-score, and accuracy. Their proposed approach achieved a high accuracy of 97%, highlighting the potential of transfer learning in medical image classification and offering a reliable tool for enhancing pneumonia diagnosis in clinical settings.

[3]Singla, M., Gill, K. S., Malhotra, S., & Devliyal, S. (2024). Chest X-ray Image Pneumonia Detection using Transfer Learning and Xception CNN Model. *2024 International Conference on Intelligent Systems for Cybersecurity (ISCS)*, 03-04 May 2024, Gurugram, India. IEEE. https://doi.org/10.1109/ISCS61804.2024.10581194

Singh et al. proposed an efficient Convolutional Neural Network (CNN) model for early pneumonia detection from chest X-ray images to help reduce mortality rates. Their model achieved a test accuracy of 93.75% and a test loss of 0.2797. The study emphasized the importance of using a diverse and well-preprocessed dataset, as well as applying deep hyperparameter tuning. Evaluation metrics such as precision, recall, and F1-score confirmed the model's effectiveness in distinguishing between pneumonia and normal cases. The authors highlighted the potential applicability of the model in real-world clinical scenarios, making it a practical tool for aiding healthcare diagnosis.

[4] G. Singh, K. Guleria, and S. Sharma, "An Efficient Convolutional Neural Network Model for Pneumonia Detection in Chest X-ray Images," in *Proc. 2024 12th International Conference on Internet of Everything, Microwave, Embedded, Communication and Networks (IEMECON)*, Jaipur, India, Oct. 2024, pp. 1–6, doi: 10.1109/IEMECON62401.2024.10845994.

III. Problem statement

Many existing approaches in the field exhibit significant overfitting, limiting their generalizability and reliability in real-world medical applications. This compromises their clinical usefulness and scalability. Our research aims to address the issue of overfitting while maintaining or improving performance metrics, thereby enhancing the robustness and practical applicability of the model in medical settings.

IV. Proposed methodology

we propose that by using genetic algorithms for training cnns we run less of a risk of overfitting and the model metrics will improve.

we used vgg16 with imagenet weights as a baseline, we removed the last layer and inserted our own, then we made multiple copies of the model running parallel to each other and started mutating the weights of the last layer, after analyzing their fitness we choose the best fit model to become the new parent and repeat.

1. Dataset and Preprocessing

   We used the Chest X-Ray Pneumonia dataset from Kaggle, which contains two classes ("pneumonia" and "normal") and a total of 5,856 X-ray images collected from a pediatric medical center . The images were initially split into training and testing sets according to the Kaggle repository. To address the severe class imbalance in the training set (fewer normal cases than pneumonia), we augmented the minority class. Augmentation operations included horizontal/vertical flipping (mirroring), random rotations, brightness and contrast adjustments, scaling (expansion), and random cropping. These data augmentation strategies artificially increased the number of minority-class examples and have been shown to reduce overfitting in CNNs . All images were preprocessed to a uniform size (e.g. 224×224 pixels) and normalized using the same scaling applied during VGG16's original training on ImageNet .

2. Model Architecture

   We adopted the VGG16 convolutional neural network as the base model, initialized with pre-trained ImageNet weights. The original final fully-connected layer (which outputs 1000 classes) was removed, and replaced with a new binary classification head appropriate for two classes (pneumonia vs. normal). Specifically, we added a dense output layer with two neurons (and softmax activation) [or one neuron with sigmoid activation], along with any necessary dropout or regularization. All other layers (all convolutional and intermediate layers of VGG16) were frozen to preserve their pre-trained feature extraction, so that during training only the new final layer's weights would be adjusted . In this way, the CNN served as a 3 f ixed feature extractor and we trained only the added layer for the target task .

3. Genetic Algorithm Optimization

   The core of our approach was to train the final layer's weights using a genetic algorithm (GA) instead of standard backpropagation. We maintained a population of 10 candidate models per generation. In each generation, the population was created by cloning the current "parent" model and then mutating the cloned models' final-layer weights. Mutation was performed by adding random perturbations (sampled from a Gaussian distribution) to the current best weights of the final layer , while leaving the frozen layers unchanged. This process produces diverse offspring models: intuitively, each "ant" or "particle" in the algorithm explores a slightly different set of final-layer weights. 4 After generating the population, each model was evaluated on a fixed validation split of the training data (the same held-out subset was used consistently). We computed classification metrics – accuracy, precision, recall, and F1-score – on the validation set for each individual. These metrics served as the fitness criteria for selection. Although multiple metrics were tracked, models were primarily ranked by their ability to correctly classify both classes (we used overall F1-score and accuracy as the main selection criteria). As in standard GA practice, the individual with the highest fitness (best performance) was chosen as the parent for the next generation. Over successive generations, the GA "evolves" better weight configurations. We repeated this 1 cycle for three generations. In summary, each generation consisted of (1) generating 10 new models by perturbing the current best final-layer weights, (2) evaluating all models on validation accuracy/recall/precision/F1, and (3) selecting the top model as the new parent. These evolutionary steps (selection, mutation, inheritance) are analogous to natural selection in GA and have been shown to find high-quality solutions over generations .

4. Comparative Optimization Techniques

   For comparison, we also trained the final layer using standard and alternative optimization methods. First, we implemented Adam (Adaptive Moment Estimation) as a baseline optimizer: using backpropagation on the same frozen VGG16 network, only the final layer's weights were tuned by Adam with default hyperparameters . In addition, we developed custom code to optimize the final layer with several nature-inspired algorithms:

- Ant Colony Optimization (ACO):

  In this approach the final-layer weight vector is treated like a "path" through the solution space. Artificial "ants" stochastically explore weight configurations, depositing virtual pheromones on promising weights. Over iterations, the algorithm probabilistically favors weights that lead to better classification performance .

- Grey Wolf Optimizer (GWO):

  This metaheuristic models the hunting hierarchy of grey wolves. Candidate solutions (wolves) are ranked as alpha, beta, delta, etc., and update their positions in weight space by tracking the leading wolves. The best three wolves guide the search each iteration, driving the pack toward optimal final-layer weights .

- Particle Swarm Optimization (PSO):

  Here a "swarm" of particle solutions (sets of weights) moves through the search space. Each particle updates its velocity and position based on its own best known and the swarm's global best-known solutions. In effect, particles "fly" toward high-performing weight configurations, which works without any gradient information .

- Bayesian Optimization:

  We also applied Bayesian optimization on the final layer. This method treats the validation performance as a black-box function of the weights, models it with a Gaussian process, and iteratively selects weight updates that maximize expected improvement. Bayesian tuning has been shown to achieve near-expert performance in optimizing deep learning hyperparameters .

Each of the above optimizers was run separately on the final layer (with the same frozen VGG16 features), using equivalent initialization and training data. The fixed validation split was used consistently across all methods to ensure fair comparison of their optimization performance.

5. Training and Evaluation Procedure

All experiments used a fixed train/validation split (for example, 80% of the data for training and 20% held out for validation). During GA optimization and all comparative methods, only the designated training portion (augmented as described) was used for learning, and the held-out validation portion was used to evaluate fitness and performance. Training proceeded for each approach until convergence on the fixed validation set or until the GA completed 3 generations (for GA) or a set number of iterations. Performance 2 was measured in terms of the chosen metrics (accuracy, precision, recall, F1), but no results are reported here. By keeping the convolutional base frozen and only tuning the final classifier layer, our methodology isolates the effect of each optimization strategy on overfitting and classification performance.

V. Conclusion

The comparative analysis highlights the significant impact of optimization algorithms on improving the performance of deep learning models for medical image classification. While the baseline CNN model achieved poor results with only 49.96% accuracy, the use of optimization techniques led to substantial performance boosts. Among all methods, the PSO-Optimized model achieved the highest accuracy (97.43%) and excellent precision, recall, and F1-score (0.98, 0.97, and 0.97 respectively), demonstrating its strength in fine-tuning model weights. Other approaches such as GA-Optimized Hybrid, ACO, and Bayesian Optimization also showed competitive results, each offering unique advantages like architecture balancing, path optimization, and hyperparameter efficiency. This clearly indicates that integrating nature-inspired optimization methods significantly enhances classification performance and can be a crucial step toward more accurate and reliable medical diagnosis systems.

TABLE I.

Model/Optimization	Accuracy	Precision	Recall	F1-Score	Key Advantages
Baseline CNN before training	49.96%	0.49	1.0000	...	
Baseline CNN after training	93.07%	0.93	0.92	...	Simple implementation
GA-(DEAP)	95.87%	0.99	0.93	0.96	Best lesion detection + classification
GA-Optimized PyGad	96.57%	0.96	0.97	0.97	Balanced architecture optimization
Grey Wolf Optimizer (GWO)	96.41%	0.98	0.94	0.96	Effective feature selection
Ant Colony (ACO)	96.65%	0.97	0.96	0.97	Good for path optimization
Particle Swarm (PSO)	97.43%	0.98	0.97	0.97	Weight tuning
Bayesian Optimization	96.41%	0.98	0.95	0.96	Hyperparameter search efficiency

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